INCH-POUND
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PERFORMANCE SPECIFICATION

GASKET MATERIAL, RUBBER, 50 DUROMETER HARDNESS (MAXIMUM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for synthetic rubber materials for gaskets of 50 durometer hardness (maximum).

1.2 Classification. The synthetic rubber is furnished in the following classes and grades, as specified (see 3.2 and 6.2):

a. Class I - Oil resistant (chloroprene rubber).

b. Class II - Non-oil resistant (butadiene-styrene rubber).

c. Class III - Fuel resistant (butadiene-acrylonitrile rubber).

(1) Grade A: Baseline Class III material (see 6.10), tensile 1500 pounds per square inch (psi) and durometer 50 or less. (see table II)

(2) Grade B: Tensile 2000 psi and durometer 40-46. (see table II)

(3) Grade C: Tensile 1700 psi and durometer 45-50. (see table II)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-190	-	Identification Marking of Rubber Products
MIL-STD-289	-	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-298	-	Visual Inspection Guide for Rubber Extruded Goods

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-STD-407	-	Visual Inspection Guide for Rubber Molded Items
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010	-	Nuclear Powered Submarine Atmosphere Control Manual

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395	-	Standard Test Methods for Rubber Property-Compression Set
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension
ASTM D471	-	Standard Test Method for Rubber Property-Effect of Liquids
ASTM D573	-	Standard Test Method for Rubber-Deterioration in an Air Oven
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D2240	-	Standard Test Method for Rubber Property-Durometer Hardness
ASTM D2565	-	Standard Practice for Xenon-Arc Exposure of Plastics Intended for Outdoor Applications

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.4), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The material shall be vulcanized synthetic rubber, which meets the applicable requirements specified herein.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Form. The synthetic rubber gasket material shall be furnished in the form specified (see 6.2). This form may be sheets, strips of rectangular cross section, shapes cut from sheets, molded shapes, or extruded shapes. Laminated form shall not be used.

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3.2.2.1 Sheet. Sheet rubber shall have smooth surfaces and shall have the thickness specified (see 6.2). Unless otherwise specified (see 6.2), the tolerances in thickness specified in table I shall apply. Unless otherwise specified (see 6.2), sheet rubber shall be furnished in rolls 36±1 inches wide and the length shall be 60±1 inches. Rolls shall consist of not more than four lengths of sheet. In no case shall the length of a sheet be less than the width.

TABLE I. Sheet rubber tolerances in width and thickness.

Width, Inches	Tolerance, Inch (Plus or Minus)	Thickness, Inch	Tolerance, Inch (Plus or Minus)
Less than 1/16	1/100	1/32 to 1/16	1/128
1/16 to 1/8, inclusive	1/64	1/16 to 1/8, inclusive	1/64
1/8 to 1/2, inclusive	1/32	1/8 to 1/2, inclusive	1/32
1/2 to 1, inclusive	3/64	1/2 to 1, inclusive	3/64
1 to 2, inclusive	1/16	Over 1	1/16
Over 2	3%		

3.2.2.2 Strip. Strip rubber of rectangular cross section shall have a smooth surface and shall have the cross sectional dimensions specified (see 6.2). Unless otherwise specified (see 6.2), the tolerances in thickness and width specified in table I shall apply. Unless otherwise specified (see 6.2), strip rubber shall be furnished in lengths of 11 feet.

3.2.2.3 Cut, molded, and extruded shapes. Cut, molded, and extruded shapes shall have the form, dimensions, and tolerances as specified (see 6.2).

3.3 Workmanship. The workmanship shall meet the requirements of this specification and gasket surfaces shall be free of surface voids, tears, rips, cuts, and free from all foreign matter that may affect the use of the finished product.

3.4 Identification. Material supplied under this specification shall be identified in accordance with MIL-STD-190 unless specifically not required in the gasket drawing. Class I material shall be marked with a yellow color and Class III material with a red color. No environmental color identification is necessary for Class II material. Marking shall be done with a permanent marking material (see 6.2).

3.5 Physical requirements. The rubber gasket materials shall conform to the requirements specified in table II.

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TABLE II. Physical requirements of rubber.

Material Condition	First Article	Conformance	Class I	Class II	Class III		Test Method
Initial properties:							
Tensile strength, psi, minimum	X	X	1500	1000	Grade A	1500	4.6.1
					Grade B	2000	
					Grade C	1700	
Ultimate elongation, percent, minimum	X	X	400	300	400		4.6.1
Tensile stress (modulus) psi at 100% elongation	X		As determined	As determined	As determined		4.6.1
Hardness, durometer points, maximum or range (Shore A)	X	X	50	50	Grade A	50	4.6.2
					Grade B	40-46	
					Grade C	45-50	
Specific gravity, maximum	X	X	1.60	1.40	1.40		4.6.3
Off-gassing	X		Limited or Permitted	Limited or Permitted	Limited or Permitted		4.8
Properties after oven aging:							
Tensile stress (modulus) psi at 100% elongation	X		As determined	As determined	As determined		4.6.1
Tensile strength, percent of initial, minimum	X		75	75	75		4.6.4.1
Ultimate elongation, percent of initial, minimum	X		70	70	70		4.6.4.1
Hardness, durometer points, maximum	X		65	65	65		4.6.4.2
Compression set, percent, maximum	X		30	30	30		4.6.4.3
Properties after light aging:							
Tensile strength, percent of initial, minimum	X		75	75	75		4.6.5
Ultimate elongation, percent of initial, minimum	X		70	70	70		4.6.5
Properties after low temperature aging:							
Flexibility	X		No cracks, tears, or separation	No cracks, tears, or separation	No cracks, tears, or separation		4.6.6
Properties after liquid immersions:							
Volume change, after water immersion, percent	X		0 to 7	N/A	N/A		4.6.7
Extraction, percent, maximum	X		0.5	N/A	N/A		4.6.8
Volume change, after immersion in IRM 901, percent	X		0 to 10	N/A	N/A		4.6.9
Volume change, after immersion in Reference Fuel B, percent	X		N/A	N/A	0 to 31		4.6.10

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3.6 Oven aging. Oven aging shall be in accordance with ASTM D573.

3.7 Toxicity. When evaluated in accordance with 4.7 (the Health Hazard Assessment [HHA]), the gasket material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.7 and 6.5).

3.8 Off-gassing. For gaskets used in submarines, the gasket material shall be evaluated for off-gassing in accordance with the requirements of 4.8. Based on the circumstances of use and the chemical nature of the gasket material, the Navy will determine whether off gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the gaskets shall be assigned to either the “Permitted” or “Limited” category (see 4.8 and 6.6).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When required (see 3.1, 6.2, and 6.4), all tests and examinations listed in 4.5 through 4.8 shall be conducted on samples from (or representing) the first lot of material.

4.2.1 Rejection for nonconformance. If any of the first article examinations or tests do not conform to the applicable requirements, this shall be cause for denial to proceed with the production of gaskets.

4.3 Conformance inspections. Conformance inspections shall include the examinations specified in 4.5 and the conformance tests specified in table II. Conformance inspections shall be conducted on samples from (or representing) all lots.

4.4 Sampling.

4.4.1 Lot. For the purposes of sampling, examinations, and tests, a lot shall consist of not more than one class (or grade, if applicable) of material of the same form and dimensions, produced in one plant under the same conditions, not exceeding 2500 pounds, and offered for delivery at one time.

4.4.2 Sampling for examination. Random samples measuring a unit area of material (see 6.7.3) shall be taken from each lot in accordance with MIL-STD-1916 for the examination specified in 4.5.

4.4.3 Sampling for tests. Representative samples shall be taken at random from each lot that passes the requirements of 4.5 in sufficient quantity to conduct the first article inspection or the conformance tests specified in 4.2 and 4.3. If the items are of such size or shape that test specimens cannot be prepared from them, a substitute sample shall be provided in the form of a piece, or pieces, of rubber having dimensions appropriate to the tests required. The substitute sample shall be the same material and equivalent cure as that used in the lot of finished material offered for delivery.

4.5 Examination. Each of the samples taken in accordance with 4.4.2 shall be subjected to surface examination for workmanship, marking, dimensions, and tolerances. MIL-STD-289, MIL-STD-298, and MIL-STD-407 shall be used to determine and evaluate visual defects in rubber sheet material, rubber extruded goods, and rubber molded items, respectively (see 6.7.1 and 6.7.2). Acceptance criteria shall be as specified (see 6.2).

4.6 Test methods. Unless otherwise indicated in the test method, tests shall not be conducted on the test specimens prior to a conditioning period of 4 hours at room temperature 74±4 °F. Sample preparation may be undertaken without regard to this time interval.

4.6.1 Tensile properties. Tensile strength, ultimate elongation, and tensile stress (modulus) at 100 percent elongation shall be determined in accordance with ASTM D412, Test Method A, using Die C specimens 0.080±0.010 inch thick.

4.6.2 Hardness. The hardness shall be determined in accordance with ASTM D2240 using a Type A durometer.

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4.6.3 Specific gravity. The specific gravity shall be in accordance with ASTM D792.

4.6.4 Oven aging. The specimens shall be oven-aged in accordance with ASTM D573, except the specimens shall be aged 46±¼ hours at 194±2 °F.

4.6.4.1 Tensile properties after oven aging. The tensile strength, ultimate elongation, and tensile stress (modulus) at 100 percent elongation after oven aging shall be determined by the procedure in 4.6.1.

4.6.4.2 Hardness after oven aging. The hardness after oven aging shall be determined by the procedure in 4.6.2.

4.6.4.3 Compression set after oven aging. Hot compression set after oven aging shall be determined in accordance with ASTM D395, Test Method B, except that specimens shall be clamped to 40 percent deflection during the oven aging, and the time and temperature of aging shall be as specified in 4.6.4.

4.6.5 Tensile properties after light aging. Specimens shall be light-aged in accordance with ASTM D2565, except the specimens shall be 0.080±0.010 inch thick, black panel temperature shall be 113±5 °F, and the exposure time shall be 100±½ hours. After light aging, the tensile strength and ultimate elongation shall be determined as specified in 4.6.1.

4.6.6 Flexibility. Specimens shall be low temperature flexibility tested as follows: specimens shall be die-cut to provide edges that are sharp, free of cracks, free of tears, and without abraded particles. Steel die shall be 0.5 inch by 6 inches. Specimens shall not exceed 0.080-inch thickness and shall be uniform to within ±0.010 inch. The specimens and a 1-inch diameter mandrel shall be exposed for 6 hours at -20±4 °F in a cold chamber. Without removing the specimens from the cold chamber, each cooled specimen shall be firmly held at one point on the 1-inch diameter mandrel and, using finger pressure, the specimen shall be forced around the mandrel within 5±1 seconds so that a 180-degree contact is achieved between the specimen and mandrel. The specimen shall not exhibit cracks, tears, or surface separation from being forced around the mandrel.

4.6.7 Volume change in distilled water (class I only). The volume change after immersion in distilled water shall be in accordance with ASTM D471, except that the immersion time shall be 24±¼ hours.

4.6.8 Extraction in distilled water (class I only). The percent extraction shall be in accordance with ASTM D471 using distilled water at boiling point for one hour. The ASTM D471 procedure used shall be “Procedure for Determining Mass of Soluble Matter Extracted by Liquid” and determining the percent extract by evaporating the water after specimen removal and weighing the non-volatile residue.

4.6.9 Volume change after immersion in oil (class I only). The volume change shall be in accordance with ASTM D471 after immersion in IRM 901 for 70±½ hours at 212±2 °F.

4.6.10 Volume change after immersion in reference fuel B (class III only). The volume change shall be in accordance with ASTM D471 after immersion in Reference Fuel B for 168±2 hours at 75±9 °F.

4.7 Toxicity. An HHA will be conducted to ensure conformance to 3.7, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the gasket material using data provided by the manufacturer/distributor to the NMCPHC (see 3.7 and 6.5).

4.8 Off-gassing. The gasket material shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.8 and 6.6). If the Navy determines that off gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.8). The Navy will review the off gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off gas testing (see 3.8).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended for general U.S. Navy gasket use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classification required (see 1.2).

c. First article, when required (see 3.1, 4.2, and 6.4).

d. Form of material required (see 3.2.2).

e. Dimensions and tolerances required (see 3.2.2.1, 3.2.2.2, and 3.2.2.3).

f. Shape of gaskets required (see 3.2.2.3).

g. Type of identification marking required, marking material, and color(s) for environmental resistance (see 3.4).

h. Requirements for an HHA (see 3.7, 4.7, and 6.5).

i. Off gas testing conformance (see 3.8).

j. Acceptance criteria for surface examination (see 4.5).

k. Packaging requirements (see 5.1).

l. Material Safety Data Sheet, when required (see 6.3).

6.3 Material safety data sheets. When required (see 6.2), contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.4 First article inspection. First article inspection should be performed when one or more of the following apply:

a. Upon initial offering of a material to the government by the vendor.

b. When any change in material formulation is made from previous first article test samples.

c. When any change in manufacturing processing cure or conditions (for example; temperature, cure time, or cure pressure) are made from previous first article test samples.

d. When the product being offered is manufactured at a plant different than that of previous first article test samples.

e. When required by the Naval Sea Systems Command.

6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

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6.6 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off gas testing. If off gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.7 Definitions.

6.7.1 Major defect. A major defect is a defect that is likely to result in failure, or to materially reduce the usability of the rubber material for its intended purpose.

6.7.2 Minor defect. A minor defect is a defect that is not likely to materially reduce the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

6.7.3 Unit area of material. A unit area of material is one square foot, regardless of thickness.

6.8 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.9 Subject term (key word) listing.

Molded shapes

Sheets

Strips

Vulcanized synthetic

6.10 Grade additions. Through revision G of the specification, three classes of gaskets existed without any grades within Class III. In revision H, three grades were added to Class III. Class III, Grade A of the latest revision is the same material as the Class III material of revision G of the specification.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MR
Navy – SH
Air Force – 11
DLA – IS

Preparing activity:
Navy – SH
(Project 5330-2015-003)

Review activities:
Army – GL
Navy – AS, MC, OS

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